

PHASE 7: PROJECT DOCUMENTATION PHASE

1. SRS Specification & Operational Scenarios Formulation [1]

This section tracks the structured user paths and interaction modes compiled during system documentation, verifying that the web presentation tier can handle various user storytelling requests smoothly.

2. Detailed Production User Workflow Scenarios [1]

Scenario 1: Standard Visual Comic Creation Pipeline [1]

User Action Path: A casual user opens the dashboard via Google Chrome, enters a raw plot paragraph: "A brave fox exploring an enchanted forest", enters the name "Leo", selects a "Dramatic" tone, picks a "Realistic" art style, and clicks the Generate Comic button

Internal Engine Operations: The FastAPI backend gathers the parameters and calls the model outline agent. Gemini Flash structures a 5-panel outline schema. Next, the creative writing agent scripts dialogue tracks, the diffusion manager downloads matching background graphics, and the matrix assembler combines text layers with file paths to load the preview display interface.

System Output State: The view layer renders comic_preview.html panel-by-panel, displaying the full visual comic book layout sequentially alongside tailored character dialogues.

Scenario 2: Mood Aesthetic Modifier Change Iterations [1]

User Action Path: The user reviews the output page but decides the comic requires a classic comic-book style. They return to the index form input page, select "Funny" as the tone, choose "Comic Book" as the art style preference, and click submit.

Internal Engine Operations: The backend controller catches the updated parameters and flushes the old image cache directories. It restarts the generation pipeline, passing new instructions that prompt the language models to write a more light-hearted story and guide the diffusion pipeline to generate classic comic-strip art.

System Output State: The view layers refresh, rendering an updated comic style featuring humorous dialogues and classic panel framing matching the user's revised creative vision.

Scenario 3: Document Vector Export Pipeline Flow [1]

User Action Path: The user reviews the final comic preview, scrolls to the bottom of the page, and clicks the green Download Your Comic as PDF button node.

Internal Engine Operations: The backend instantiates the `save_pdf()` processing hub. The engine calls native canvas drivers to automatically place headings, embed local JPEG assets onto consecutive pages, and wrap multi-line speech text fields cleanly within margins. It outputs a timestamped file configuration block into local storage disk areas and returns the file path.

System Output State: The user browser is redirected to the export success interface viewport page (`export_success.html`), confirming the download, while the printable digital comic file downloads directly onto the client's local storage storage device.